Rotational Shearing Term

March 10, 2026

Applying Rotational Shear

Solving by the method of characteristics

Considering only the part of the equation which contains the shearing term:

$$\frac{\partial E}{\partial z} = \frac{\Omega}{c} z \frac{\partial E}{\partial x} \tag{1}$$

We can solve the PDE by the **method of characteristics.** We transform the PDE into an ODE and then solve for the curves where E is constant.

$$\frac{dE}{ds} = E_x \frac{dx}{ds} + E_z \frac{dz}{ds} = 0 \tag{2}$$

$$\frac{dx}{ds} = -\frac{\Omega}{c} z \tag{3}$$

$$\frac{dE}{ds} = 0 \tag{4}$$

$$\frac{dz}{ds} = 1 \tag{5}$$

Which means that

$$z = s + const.$$

Or simply, redefining s,

$$z = s$$

Now, from equation (3),

$$\frac{dx}{dz} = -\frac{\Omega}{c}z \tag{6}$$

Solving this, we get

$$x - x0 = -\frac{\Omega}{2c}(z^2 - z_0^2) \tag{7}$$

So E depends only on the invariant $C = x + \frac{\Omega}{2c}z^2$ and the solution is:

$$E(x, z) = E_0\left(x + \frac{\Omega}{2c}z^2\right) \tag{8}$$

i.e. As z increases, the pattern in E slides in x by $\frac{\Omega}{2c}z^2$

Discretization

We know that at step z

$$v_x(z) = \frac{\Omega}{c}z$$

So at the next step,

$$v_x(z + \Delta z) = \frac{\Omega}{c}(z + \Delta z)$$

Averaging over the interval,

$$\bar{v}_x = \frac{1}{2}\Big[v_x(z) + v_x(z + \Delta z)\Big] = \frac{\Omega}{c}\left(z + \frac{\Delta z}{2}\right)$$

And so

$$\Delta x = \bar{v}_x \Delta z = \frac{\Omega}{c}\left(z + \frac{\Delta z}{2}\right)\Delta z$$

In order to apply this shift in the code, we can shift everything in x by Δx and interpolate onto the new grid.

Results

Using the method described above, after 100 passes through a cavity 100 km long, with $R_c = 200$ km, $D = 0.5$ m, and $dz = L$, we converge to the following lowest-order mode:

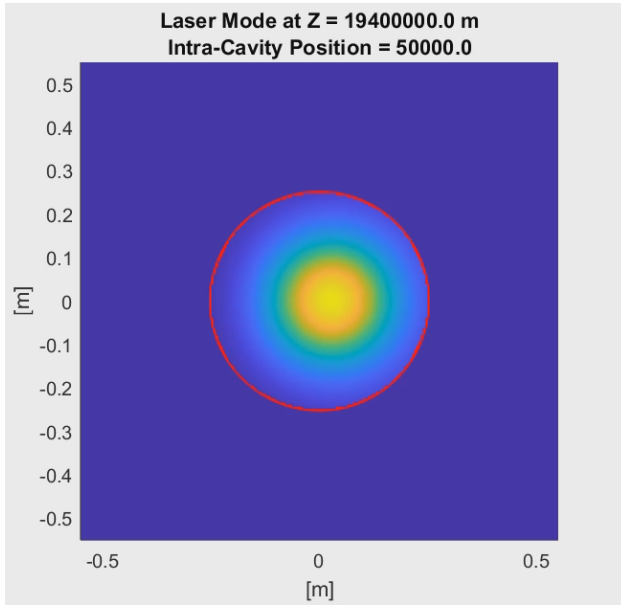

Figure 1: The mode is slightly shifted on the mirror face.

Note

Note that the method of manually shifting the mode over using the "geometric" solution

$$\Delta t = \frac{\Delta z}{c}$$

and

$$\Delta x = \Omega z \Delta t$$

does work, but it requires a relatively small dz step to return an accurate solution. For example, here is the converged solution with $dz = L$ versus the solution with a decrease in step size to $dz = L/4$.

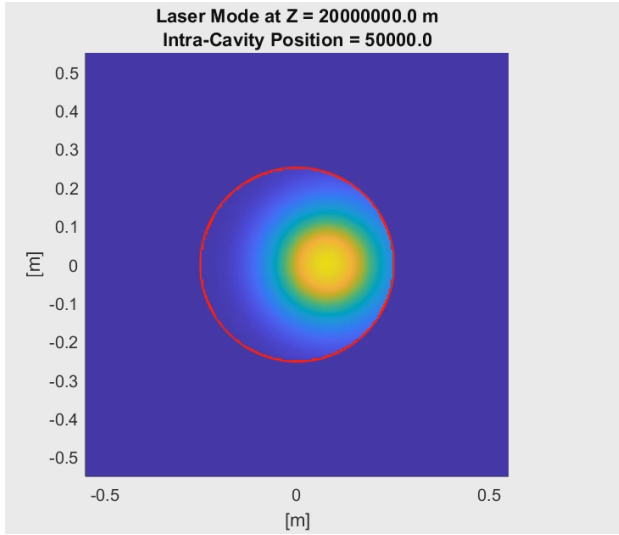

Figure 2: Converged solution with $dz = L$.

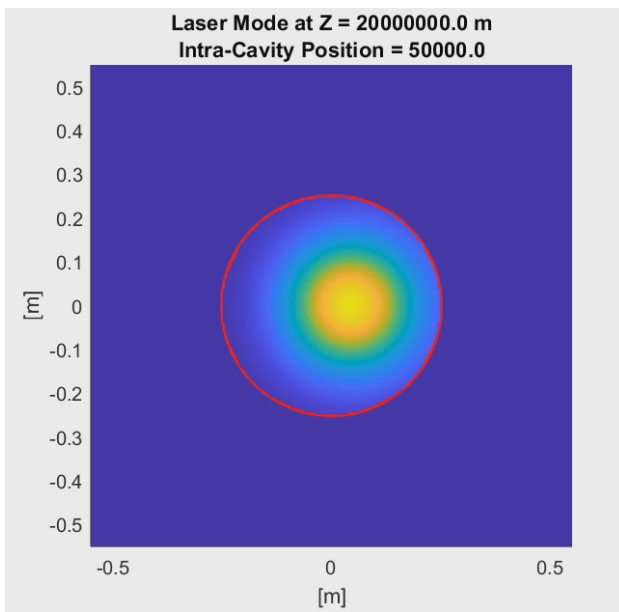

Figure 3: Converged solution with $dz = L/4$

We see that the accuracy of the solution improves with the decrease in dz. This approach is only approximate because it assumes v_x is constant over the step.

$$\frac{dA}{dZ} = \frac{\Omega}{c} Z \frac{dA}{dX}$$

$$0 \leq g_1 g_2 \leq 1$$

$$g_2 \equiv 1 - \frac{L}{R_2}$$

$$\nabla^2 E + k^2 E = 0$$

$$E(x, y, z) = A(x, y, z) e^{ikz}$$

$$\frac{1}{c}\frac{\partial A}{\partial t} + \frac{\partial A}{\partial z} = \frac{i}{2k}\nabla_\perp^2 A$$

$$\Theta_D = \frac{\lambda}{D}$$